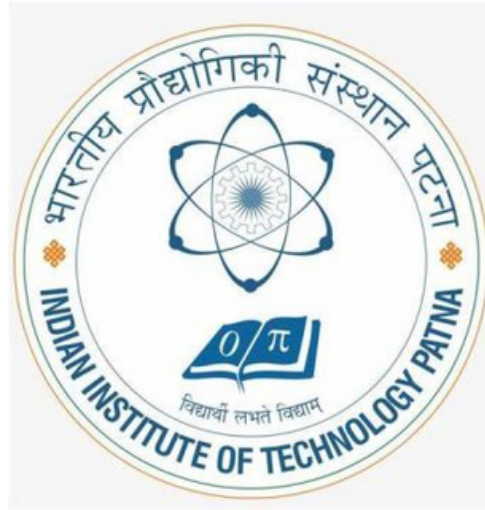


Indian Institute of Technology Patna



Group 02

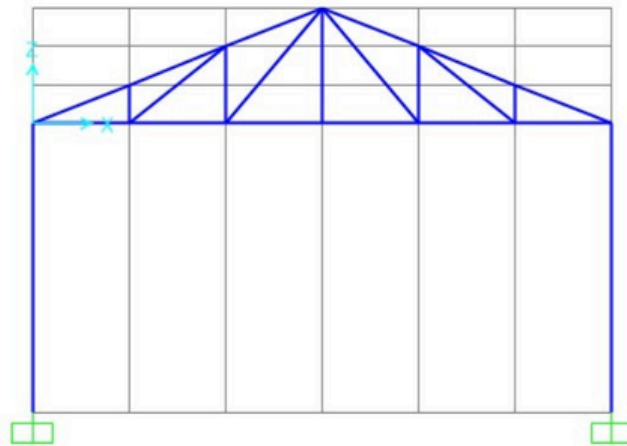
Session: 2025 - 2026

SAP2000 Analysis Report

Pratt Truss



Front View



3D View

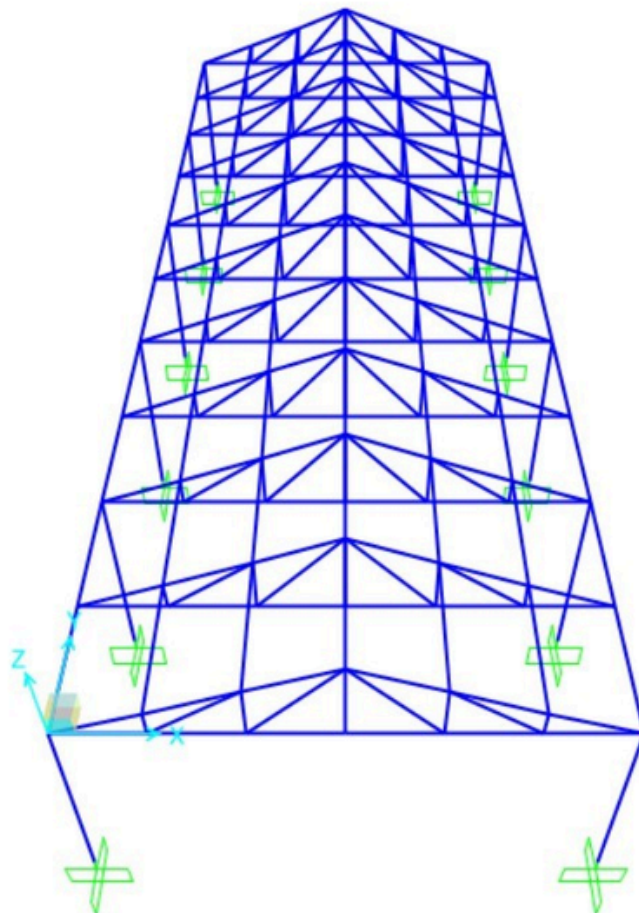


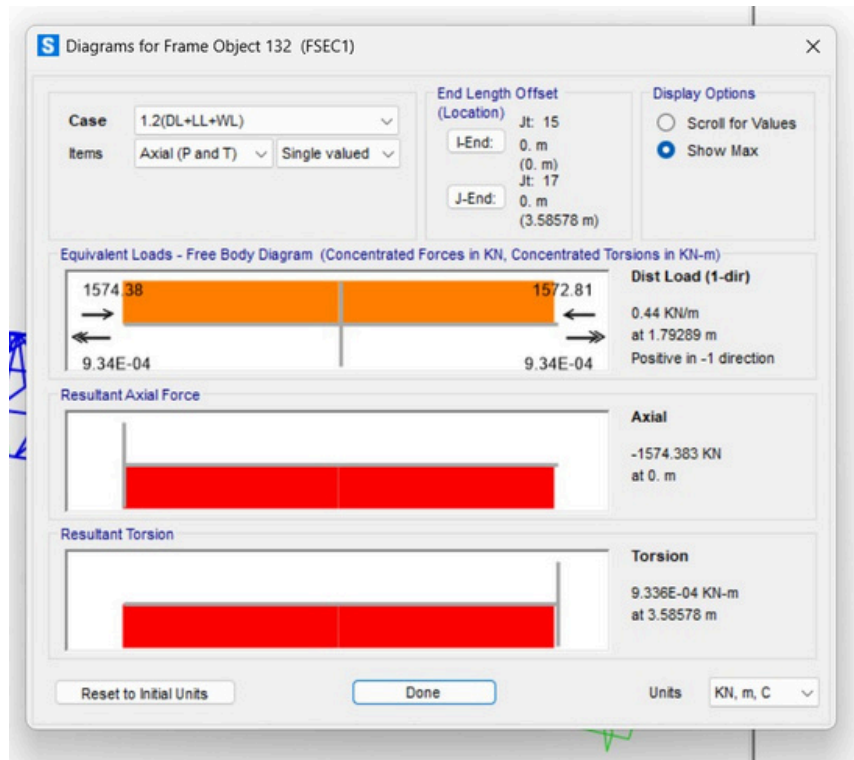
Table Forces in Critical Members

Member No.	Member	Load Type	Maximum Load (KN)	Load Case
1.	Top Chord	Axial Force	-1098.359	1.5(DL+LL)
			-1510.437	1.5(DL+WL)
			-1574.383	1.2(DL+LL+WL)
2.	Vertical	Axial Force	-222.5	1.5(DL+LL)
			-316.911	1.5(DL+WL)
			-327.306	1.2(DL+LL+WL)
3.	Diagonal	Axial Force	306.98	1.5(DL+LL)
			432.23	1.5(DL+WL)
			448.232	1.2(DL+LL+WL)
4.	Beam	Shear Force	-100.206	1.5(DL+LL)
			-130.913	1.5(DL+WL)
			-137.714	1.2(DL+LL+WL)
		Bending Moment	285.7508(kn-m)	1.5(DL+LL)
			406.2704(kn-m)	1.5(DL+WL)
			420.0579(kn-m)	1.2(DL+LL+WL)
5.	Purlin (Major)	Shear Force	-87.998	1.5(DL+LL)
			-125.545	1.5(DL+WL)
			-129.342	1.2(DL+LL+WL)
		Bending Moment	172.8284(kn-m)	1.5(DL+LL)
			-241.552(kn-m)	1.5(DL+WL)
			-250.3477(kn-m)	1.2(DL+LL+WL)
6.	Purlin (Minor)	Shear Force	-0.108	1.5(DL+LL)
			-11.545	1.5(DL+WL)
			-9.272	1.2(DL+LL+WL)
		Bending Moment	-0.3288(kn-m)	1.5(DL+LL)
			-9.946(kn-m)	1.5(DL+WL)
			-8.0663(kn-m)	1.2(DL+LL+WL)
7.	Horizontal	Axial Force	994.372	1.5(DL+LL)
			1356.182	1.5(DL+WL)
			1416.29	1.2(DL+LL+WL)
8.	Column	Axial Force	-769.066	1.5(DL+LL)
			-1040.95	1.5(DL+WL)
			-1088.91	1.2(DL+LL+WL)

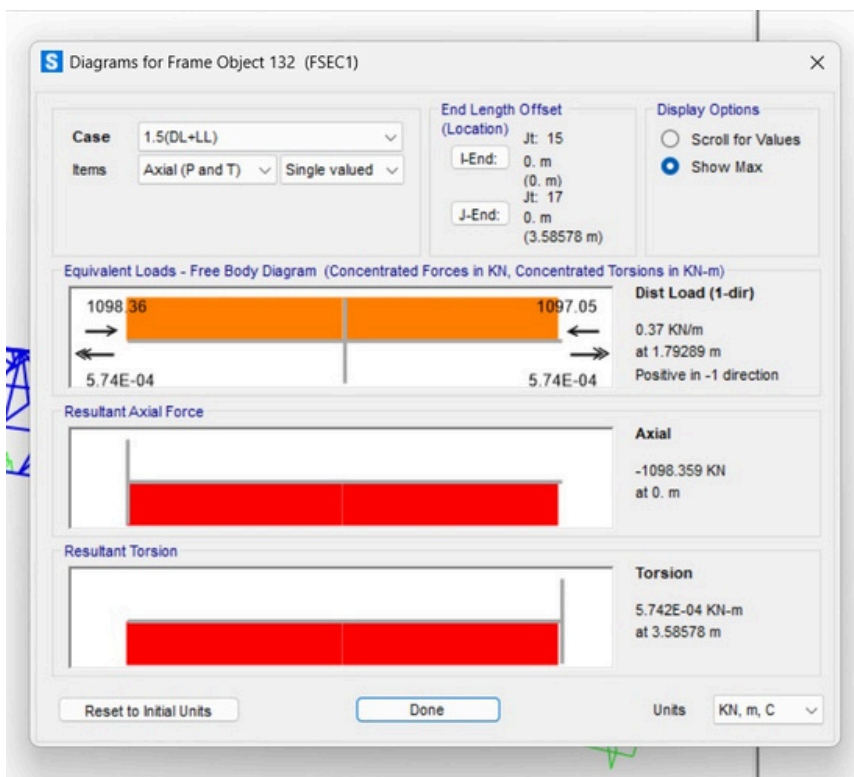
Truss Member Analysis

1. Top Chord

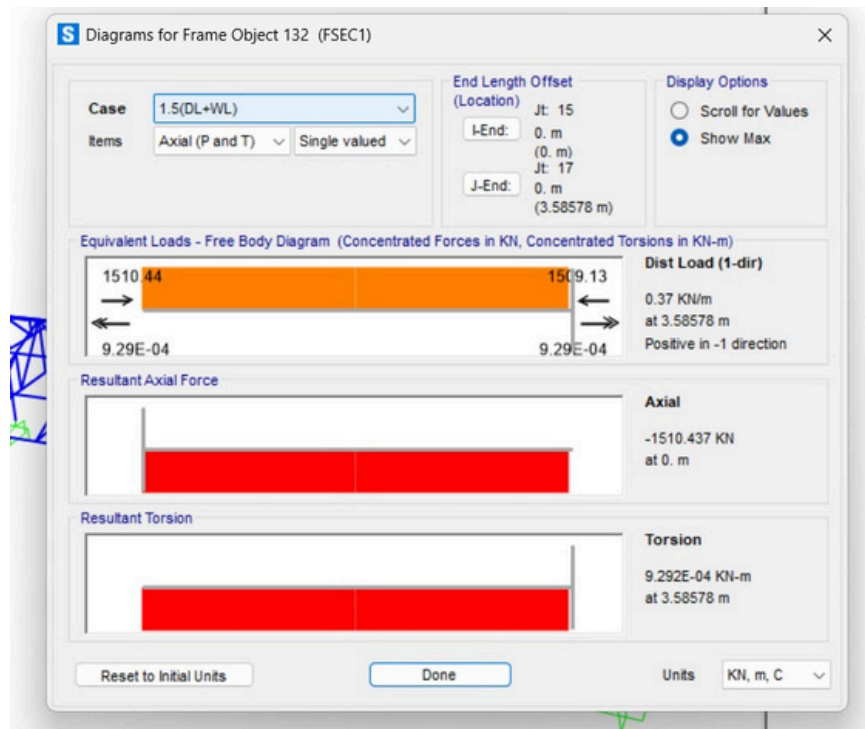
1.2(DL+LL+WL)



1.5(DL+LL)

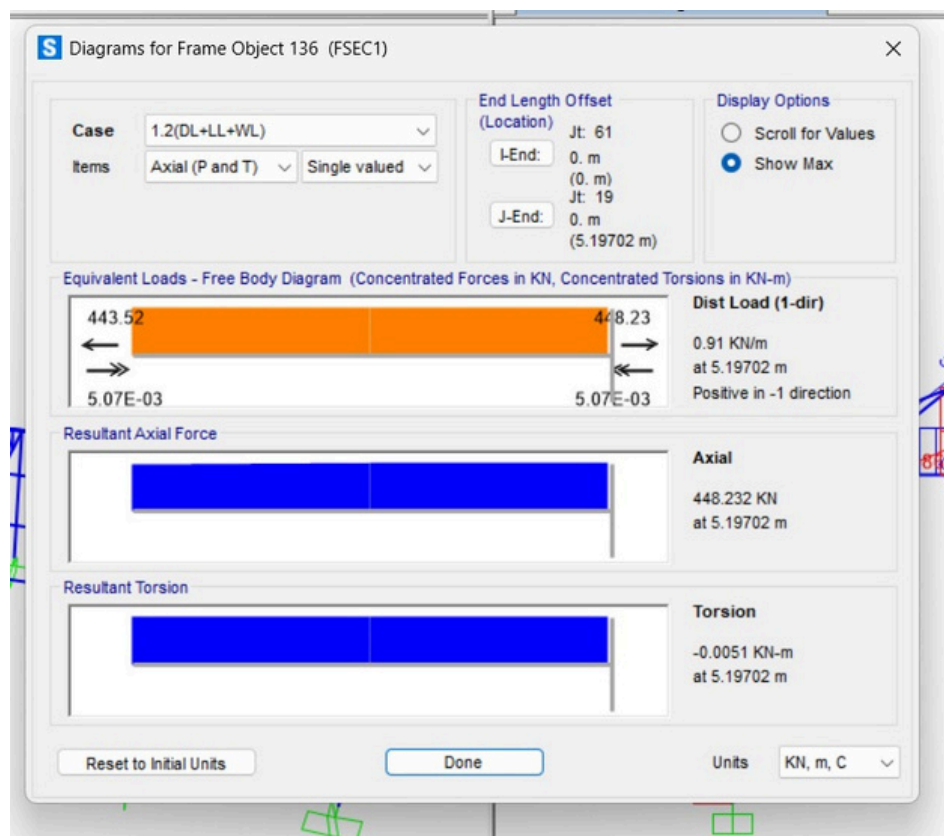


1.5(DL+WL)

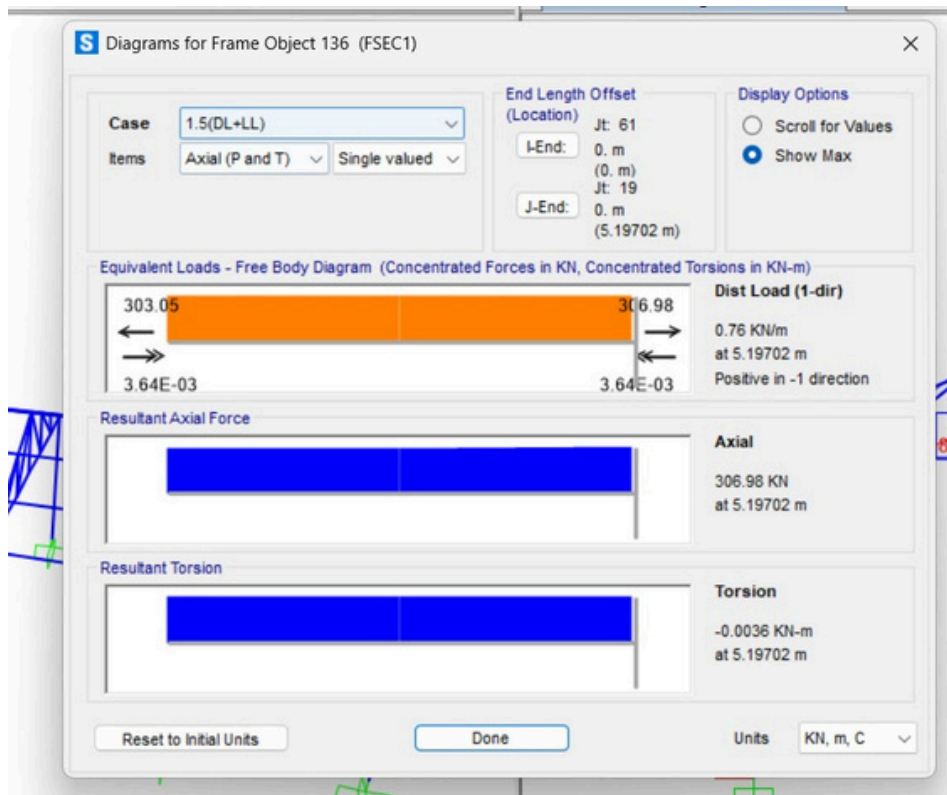


2. Diagonal Chord

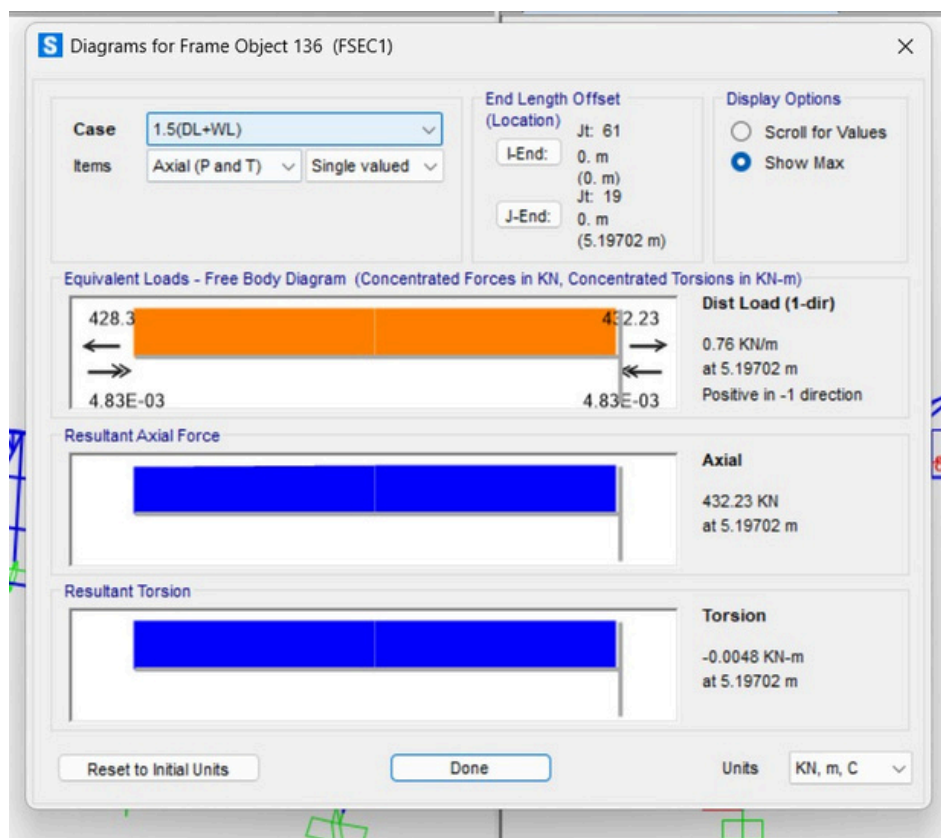
1.2(DL+LL+WL)



1.5(DL+LL)

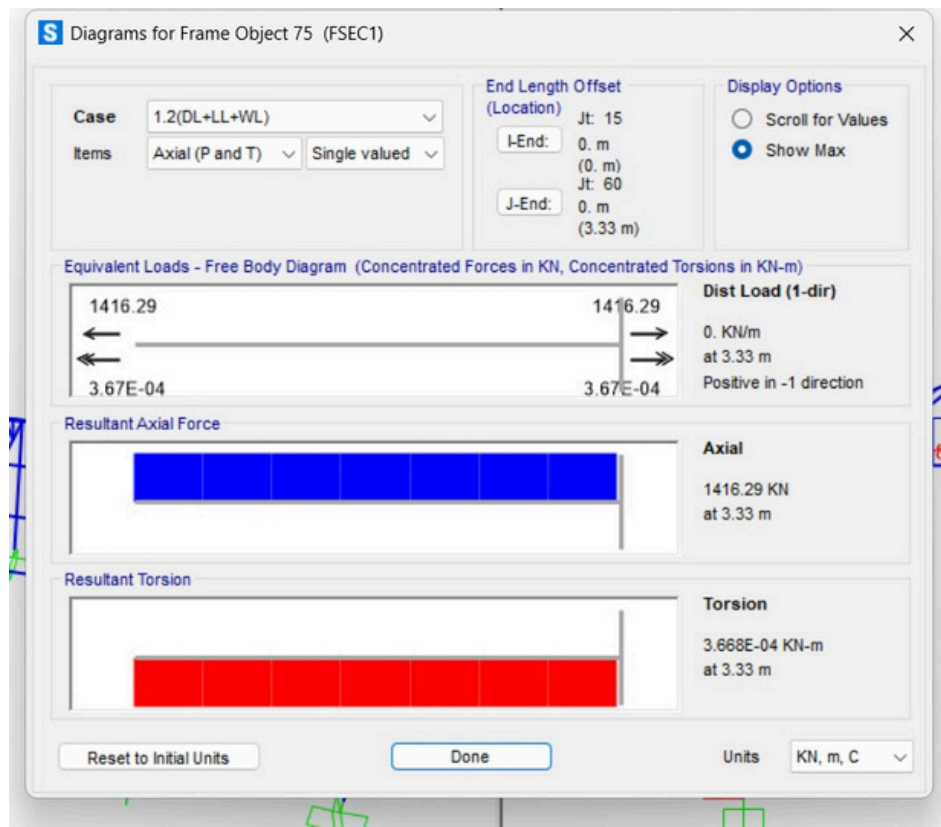


1.5(DL+WL)

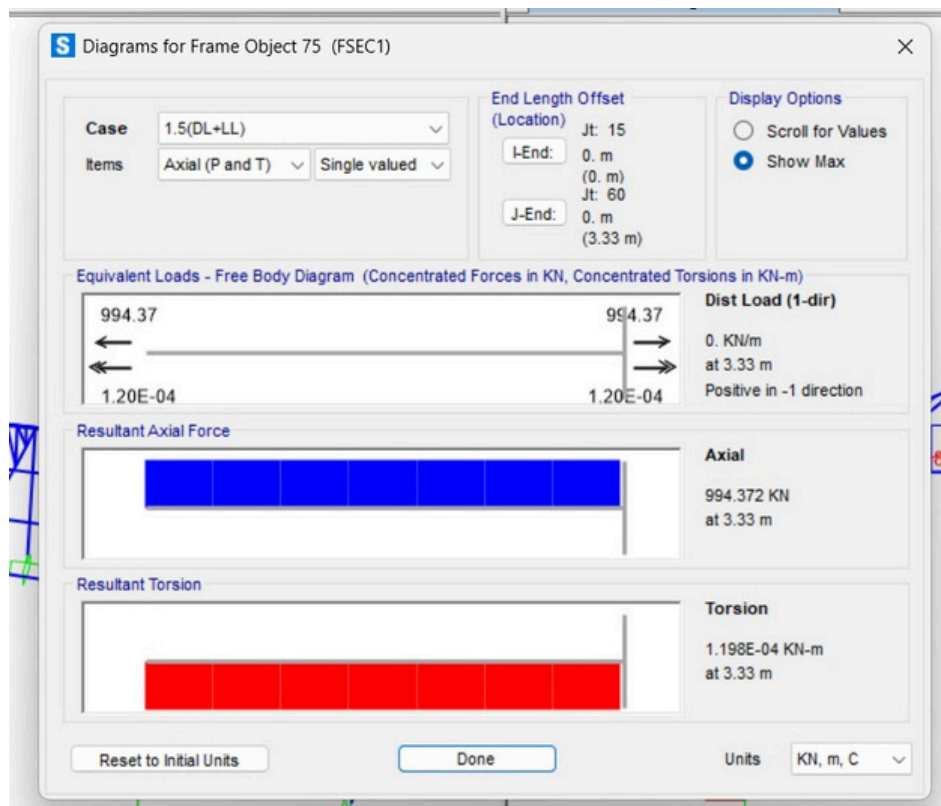


3. Horizontal Chord

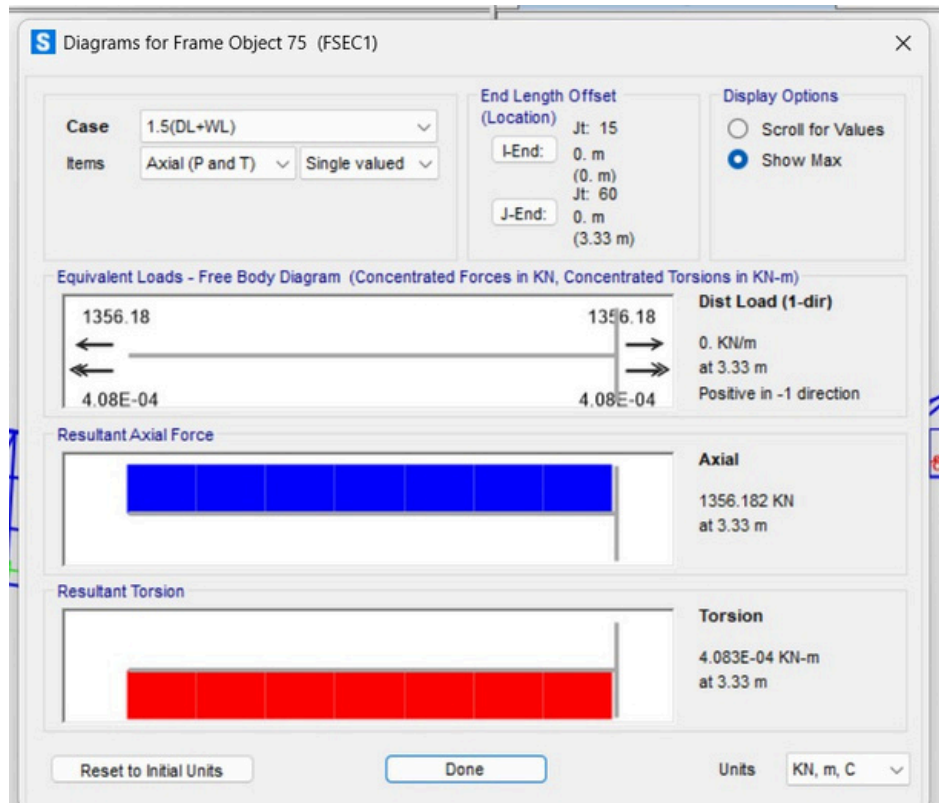
1.2(DL+LL+WL)



1.5(DL+LL)

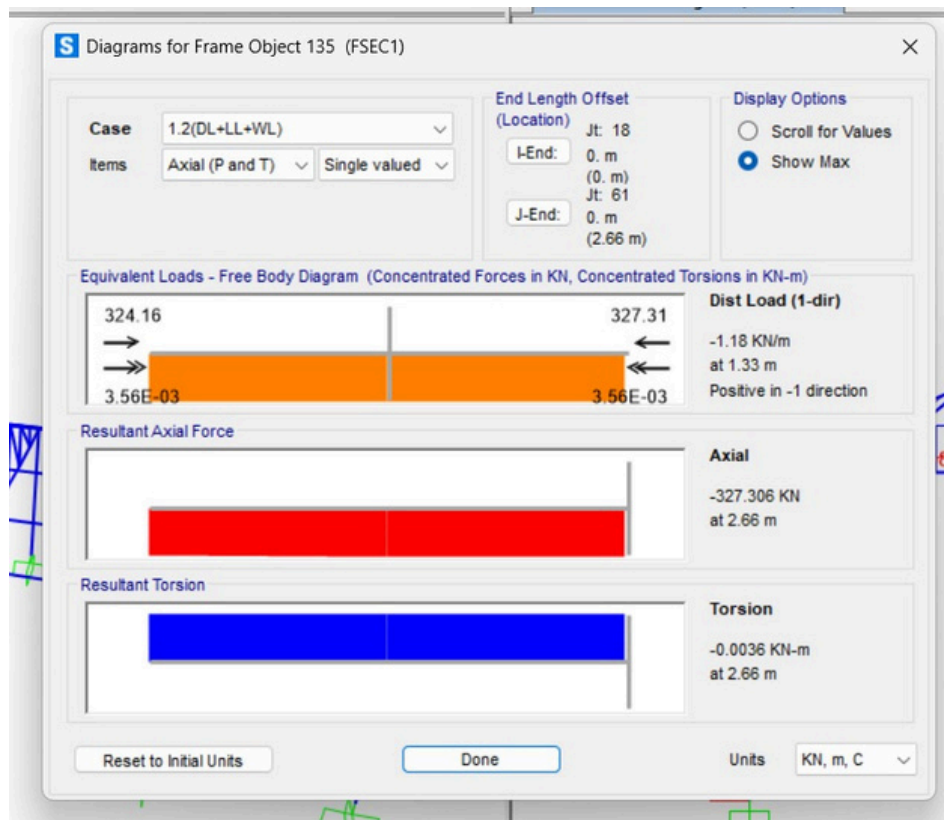


1.5(DL+WL)

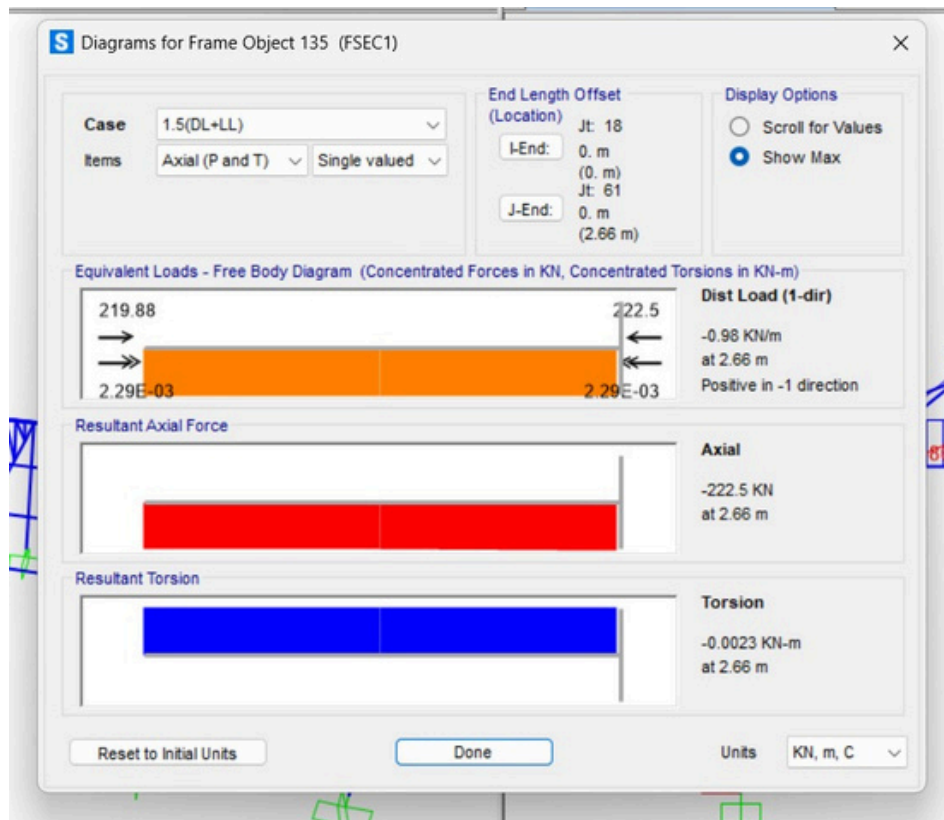


4. Vertical Chord

1.2(DL+LL+WL)



1.5(DL+LL)



1.5(DL+WL)

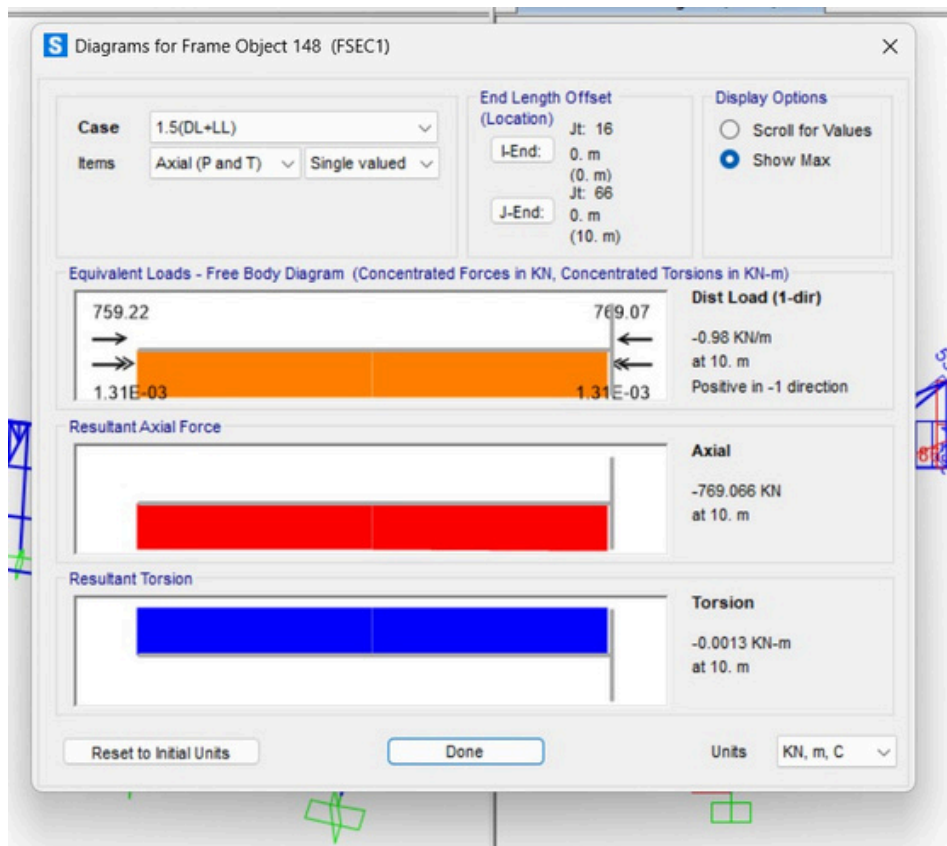


Column Analysis

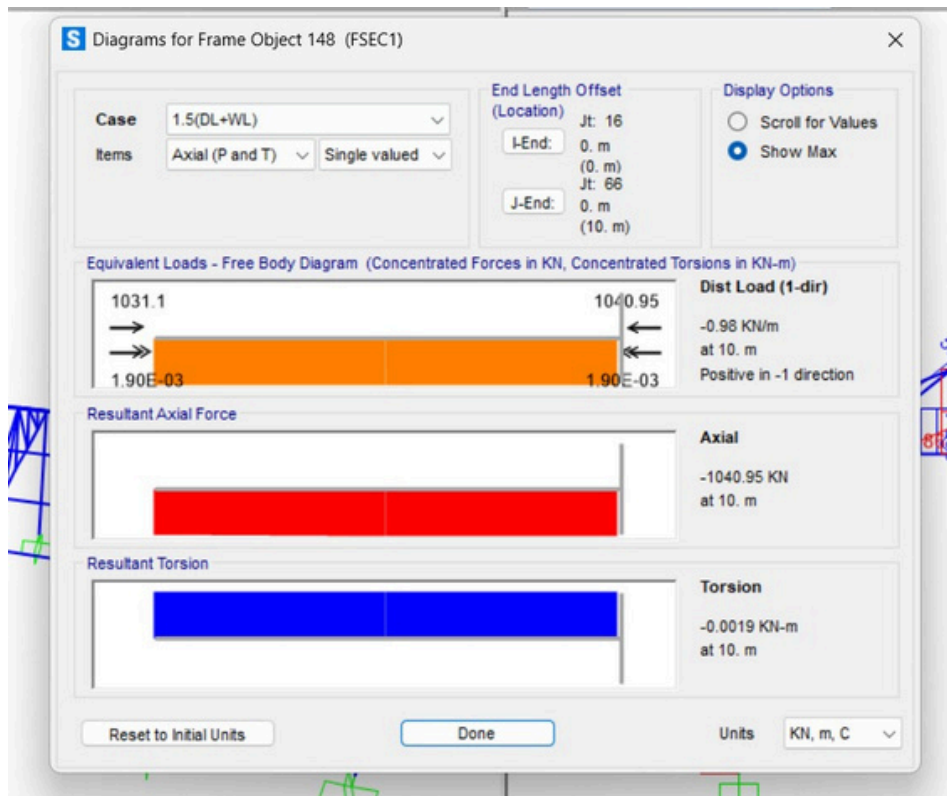
1.2(DL+LL+WL)



1.5(DL+LL)

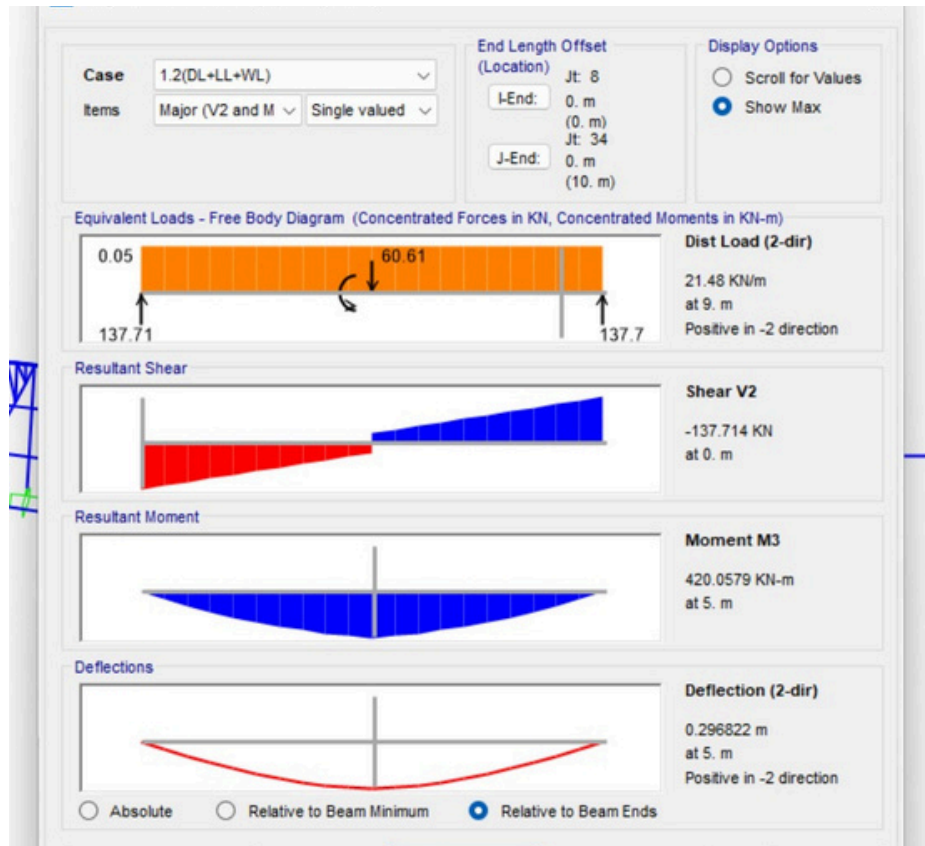


1.5(DL+WL)

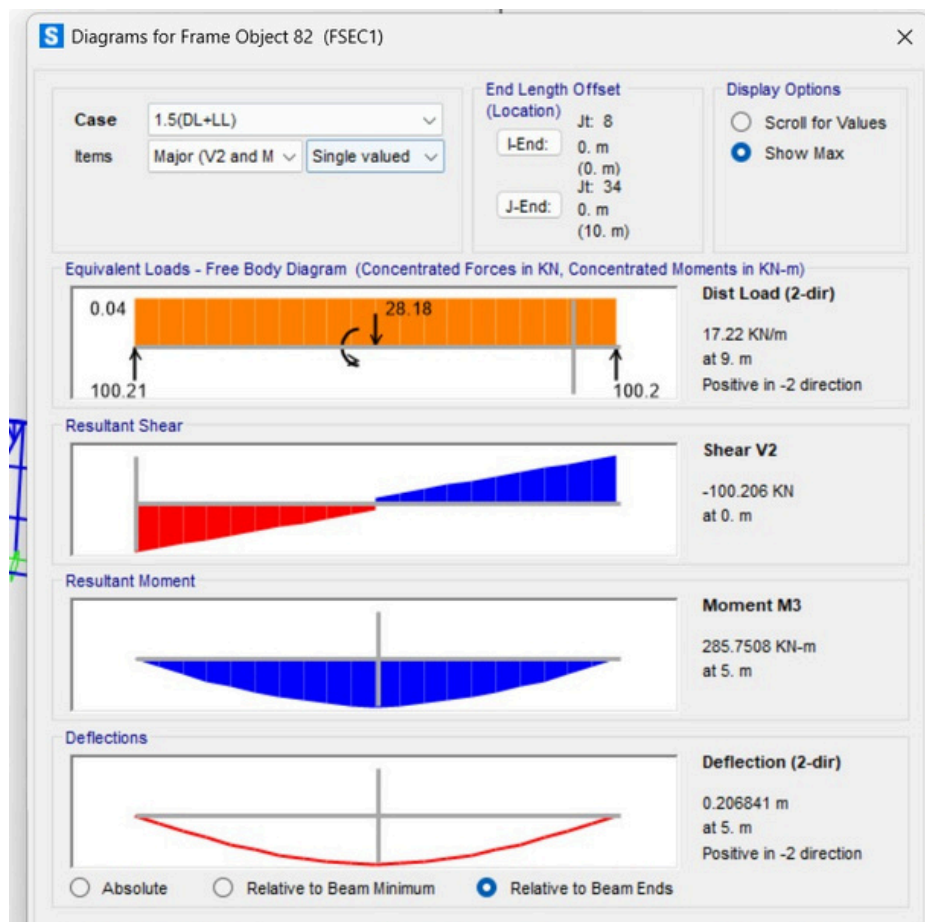


Girder Analysis

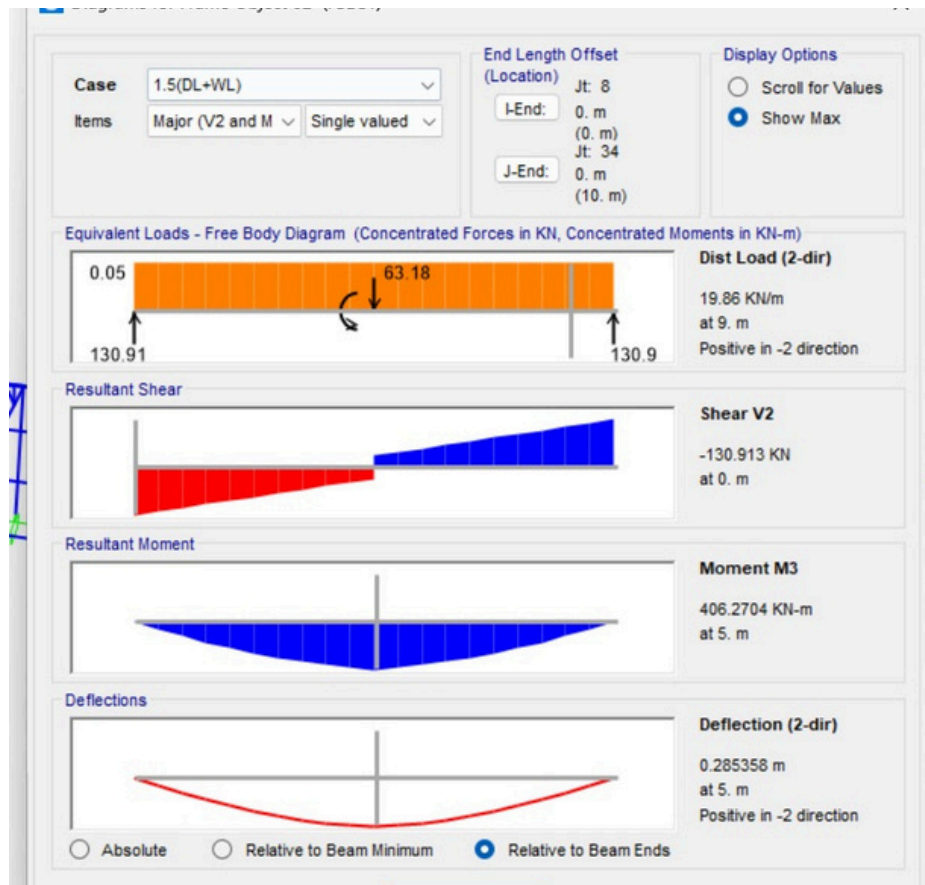
1.2(DL+LL+WL)



1.5(DL+LL)



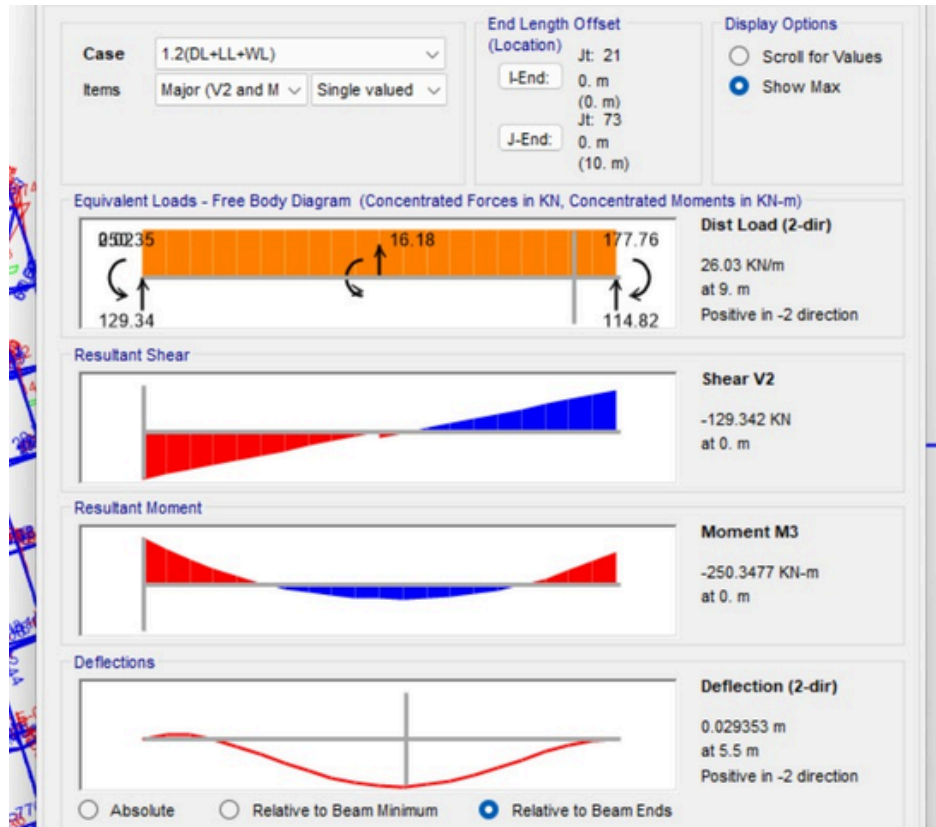
1.5(DL+WL)



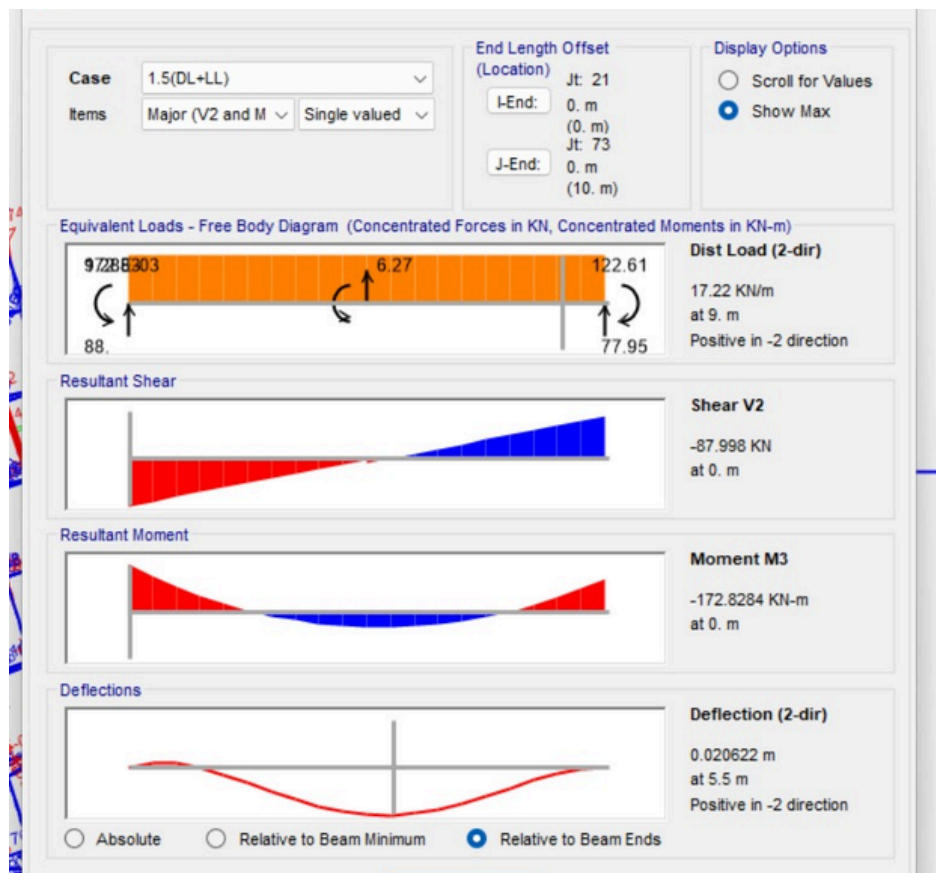
Purlin Analysis

1. Major

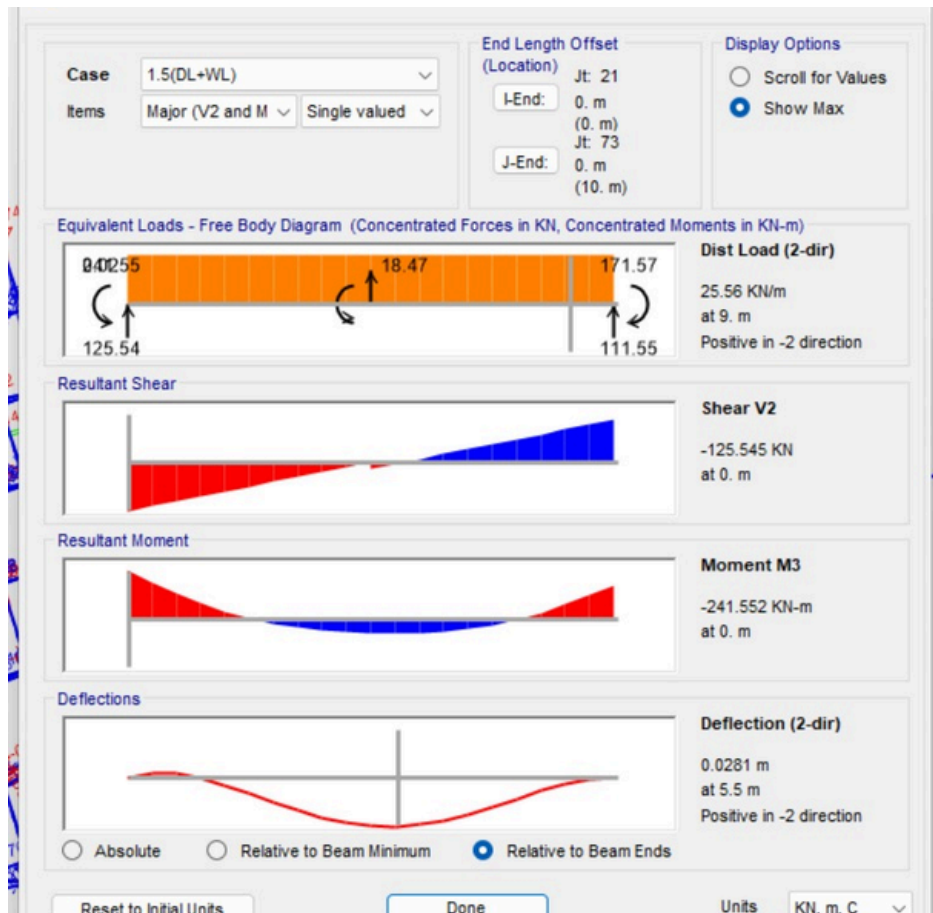
1.2(DL+LL+WL)



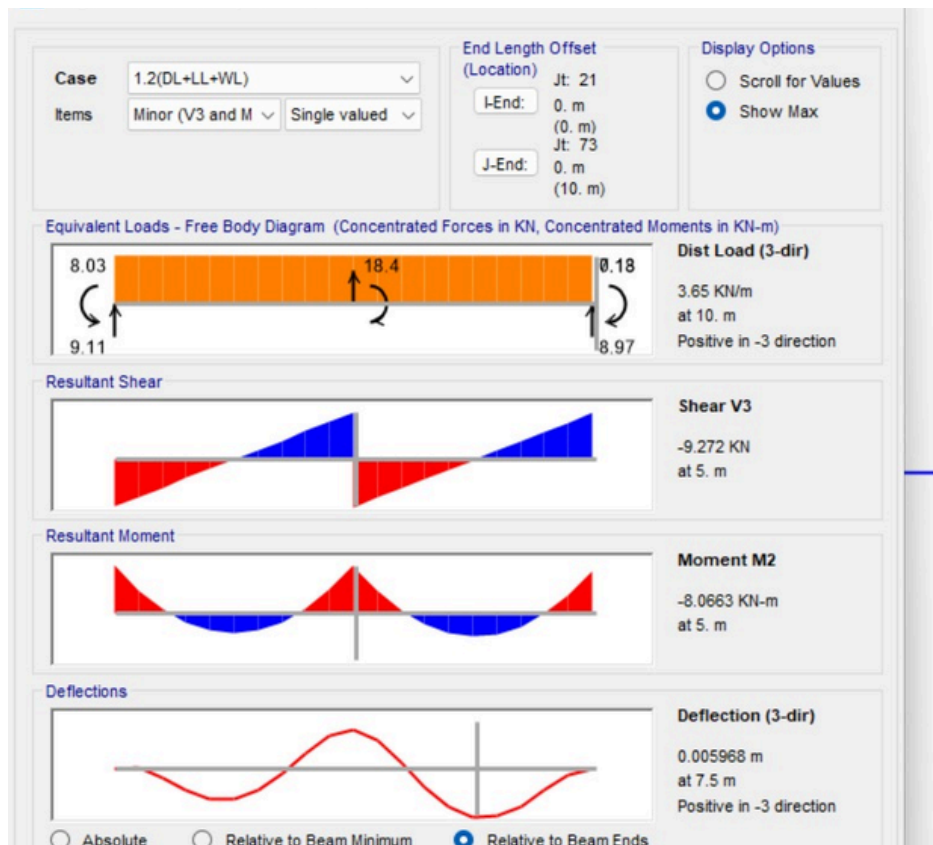
1.5(DL+LL)



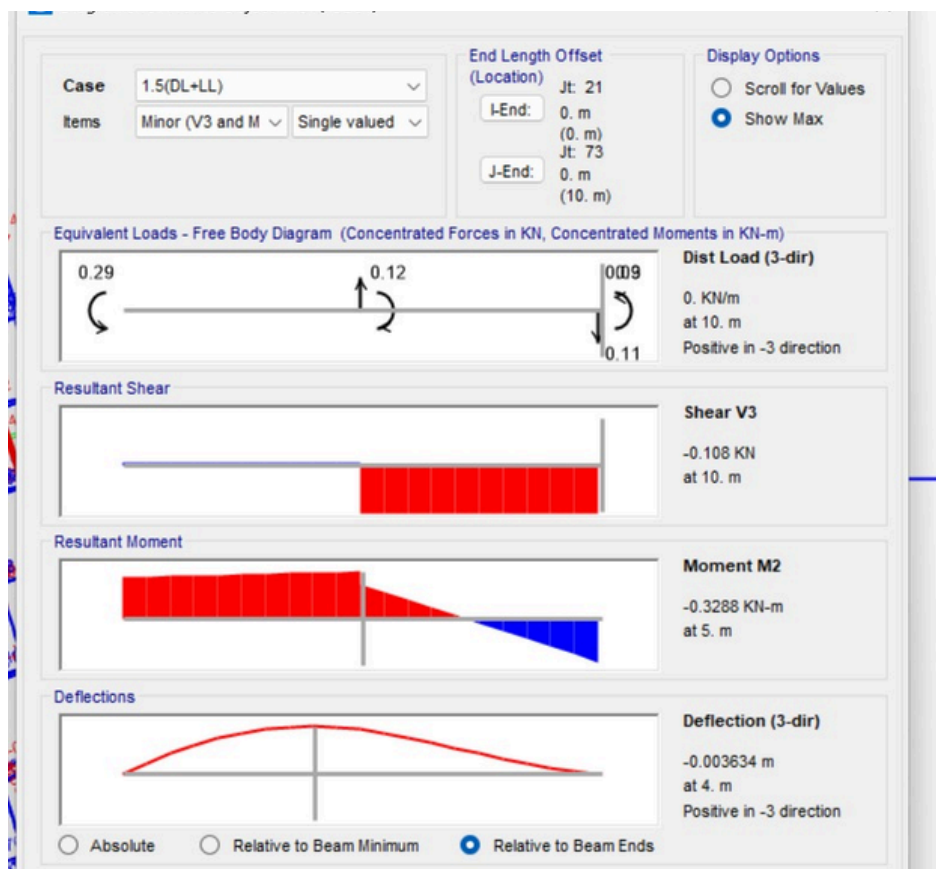
1.5(DL+WL)



2. Minor 1.2(DL+LL+WL)



1.5(DL+LL)



1.5(DL+WL)

